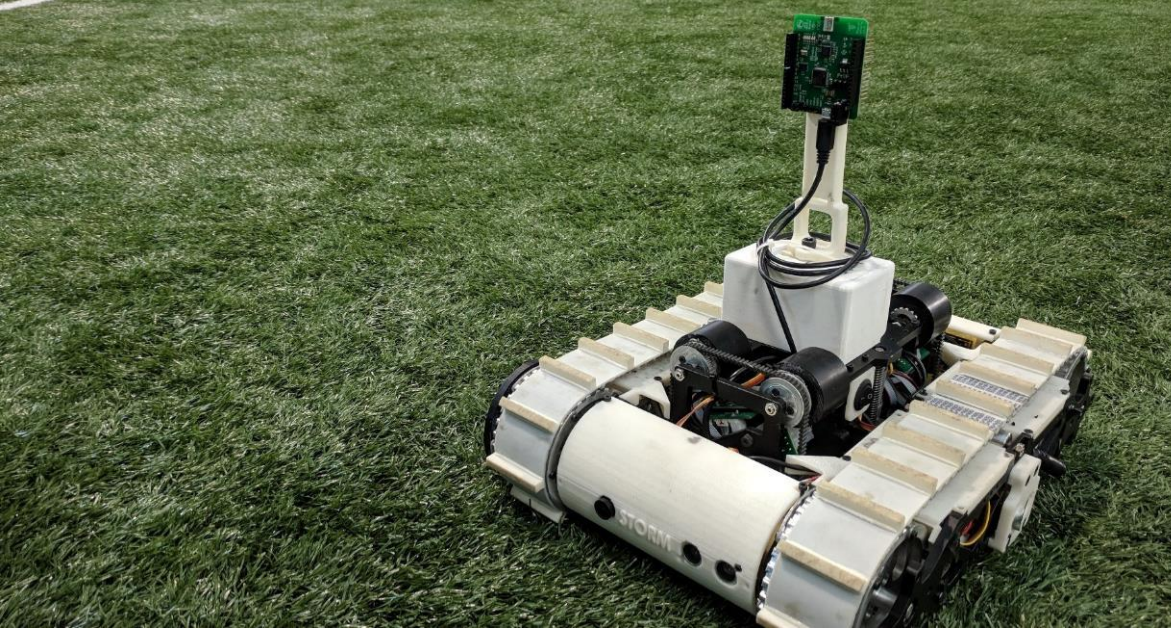




Path Planning for Pioneer P3-AT

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PROJECT DEFINITION

- **Problem:** Find an efficient path for a Pioneer P3-AT robot to navigate from its initial location to a goal in a dynamic environment
- The motion planning execution must satisfy the constraints of skid-steer wheeled robot dynamics
- This includes finding a path to the goal and maneuvering the vehicle along this path to avoid obstacles
- Maps will be given, but will change over time
- Algorithm must be adaptive to changes, without having to replan from scratch

We make various assumptions like we have perfect sensing, access to the global map at all times, the robot will not run out of battery, will get its global coordinates, moving obstacles have linear trajectories, etc, in order to focus on the motion planning aspects of this problem.



DELIVERABLES

Path Planning in Dynamic Environments (Simulation)

- Implement adaptive motion planning for a **Pioneer P3-AT** robot in **Gazebo**.
- Navigate through multiple simulated environments with varying layouts, static and dynamic obstacles with **different behaviours/trajectories**.

Performance Analysis & Optimization

- Evaluate **path efficiency**, **optimality**, and **adaptability** in dynamic scenarios.
- Compare different path-planning algorithms and refine for better real-time performance.

Real-World Deployment

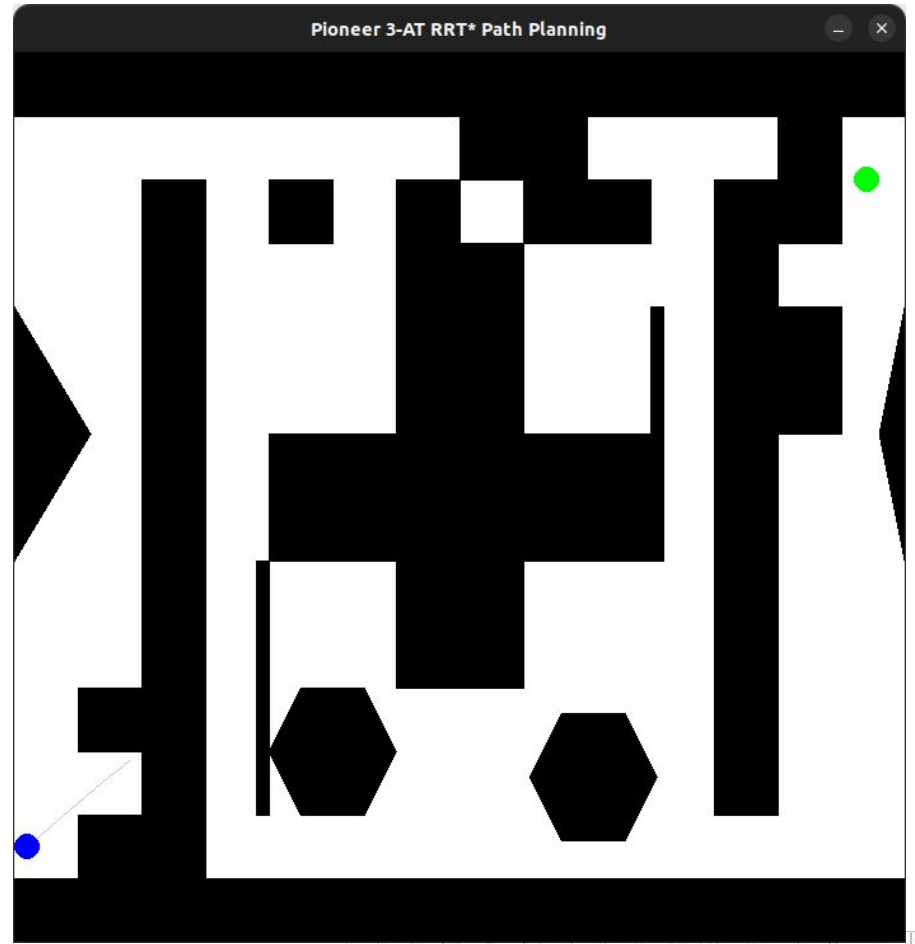
- Validate the most optimal and efficient algorithm on a **physical robot**.
- Optimize for real-world uncertainties like sensor noise/drift and wheel slippage.

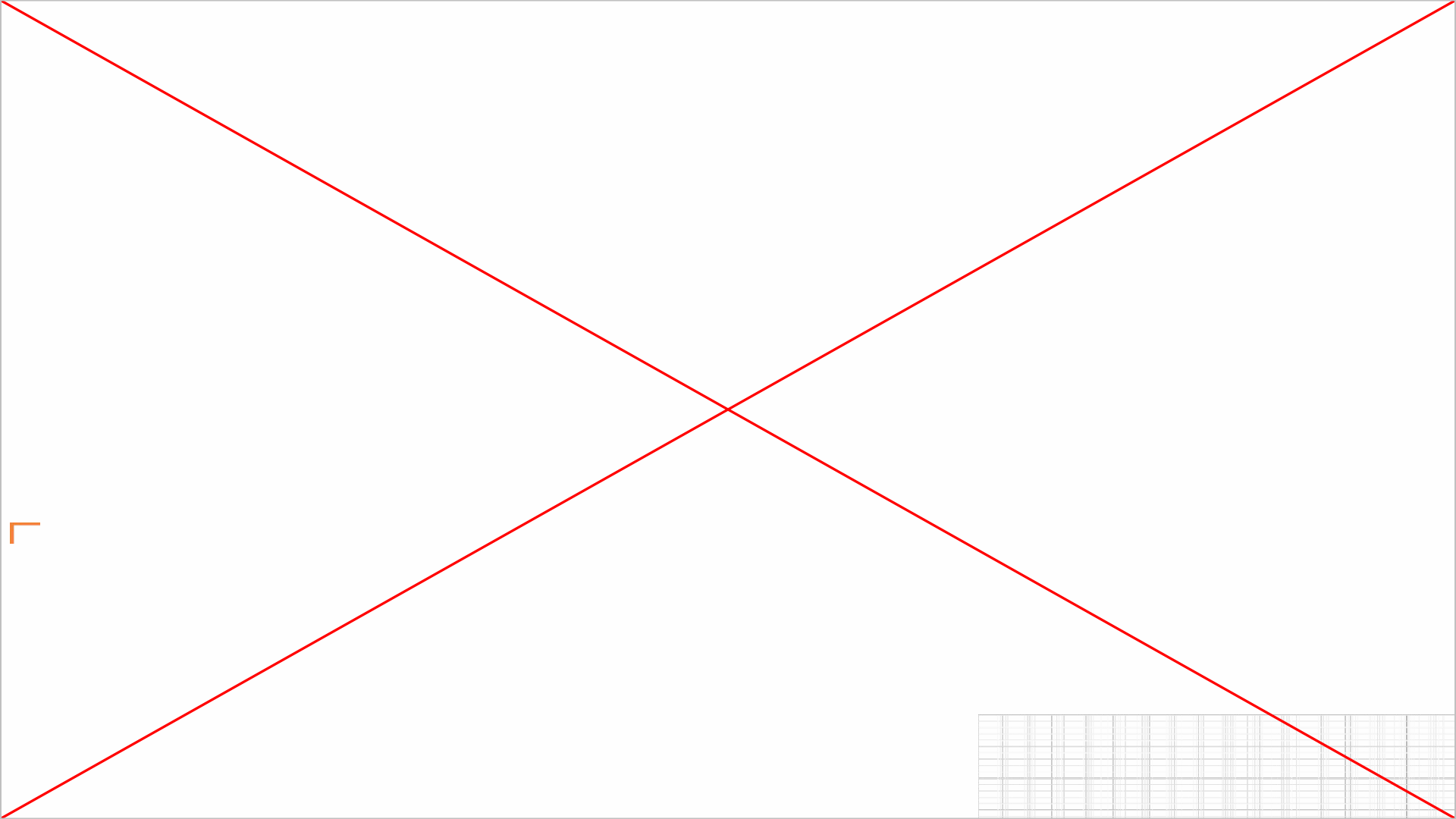
Extended deliverables:

- Extend the simulation framework to handle **multiple autonomous robots**, each with different goal positions.
- Implement a decentralized/cooperative planning strategy to **avoid collisions and optimize efficiency**.

Stage 1: Path planning with RRT*

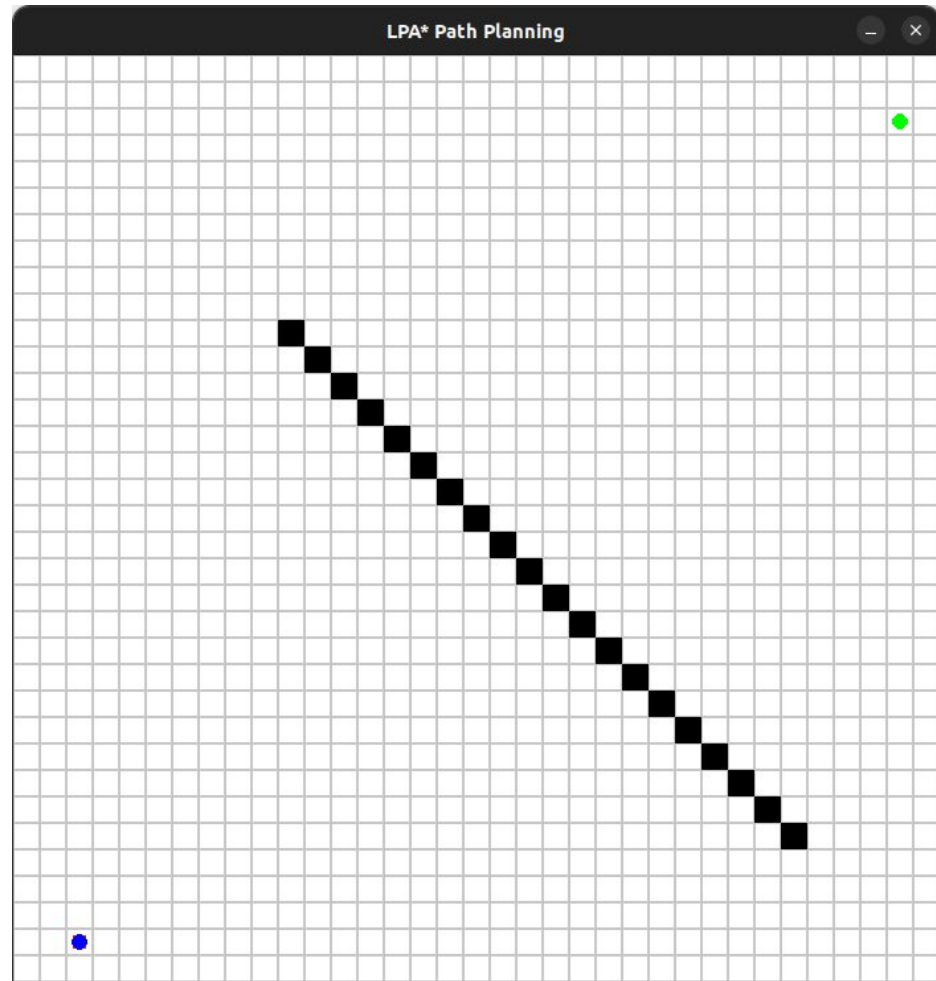
Static environment (700 * 700 px), known map, point robot

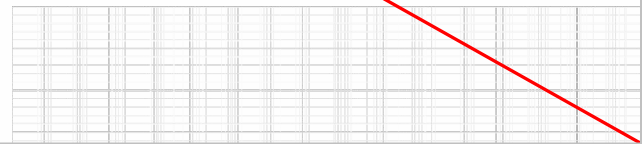
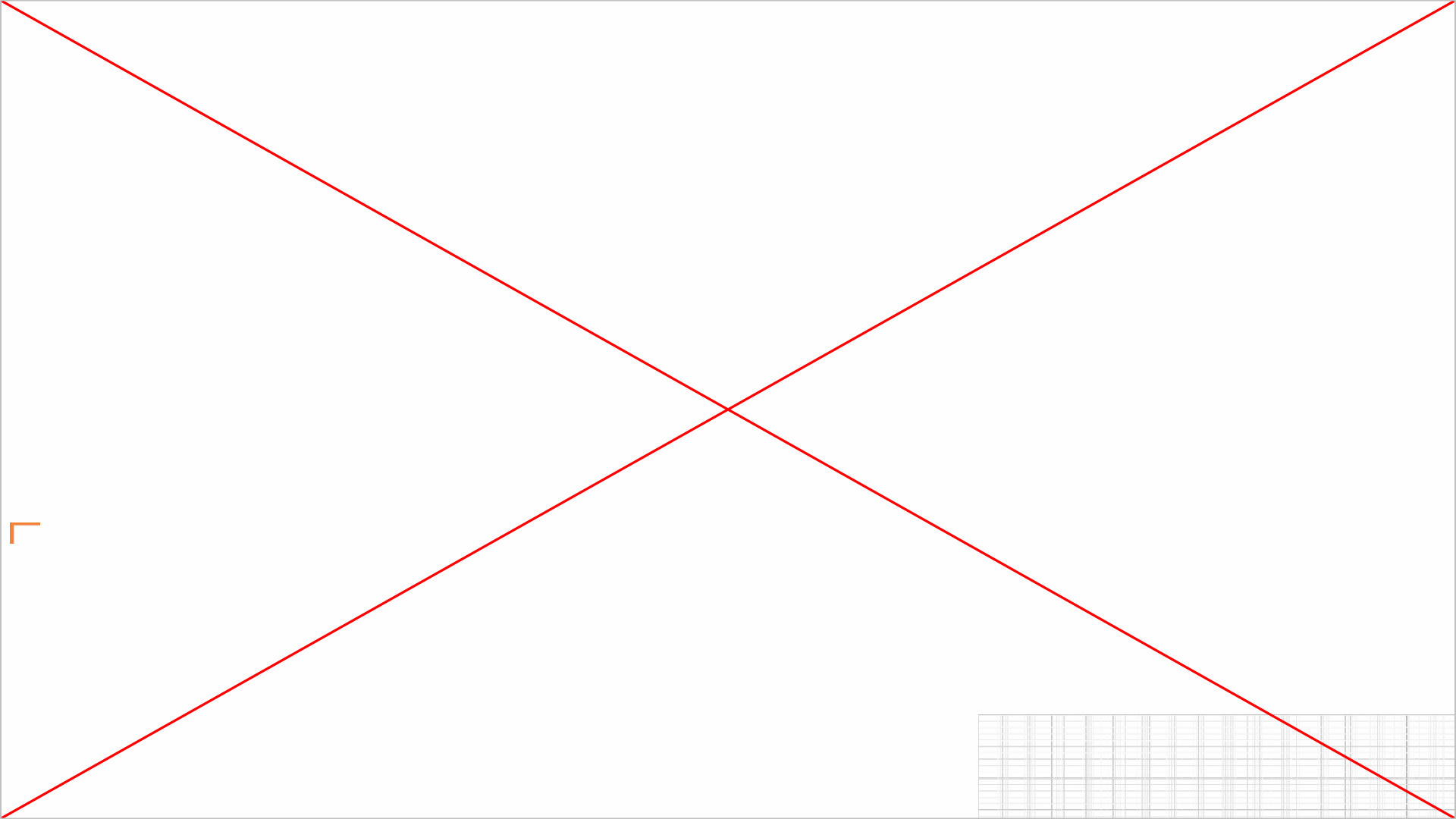




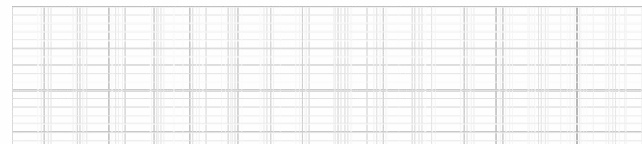
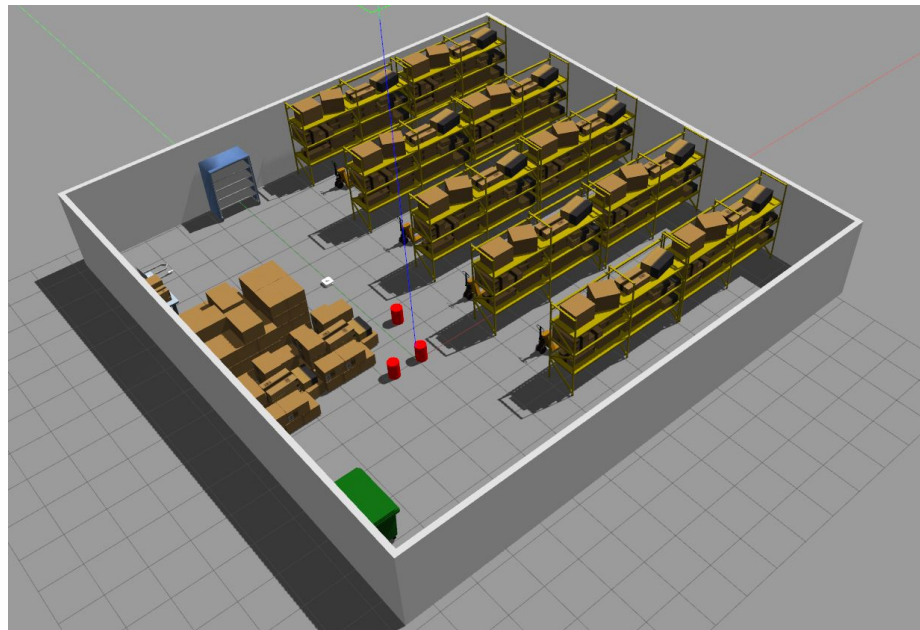
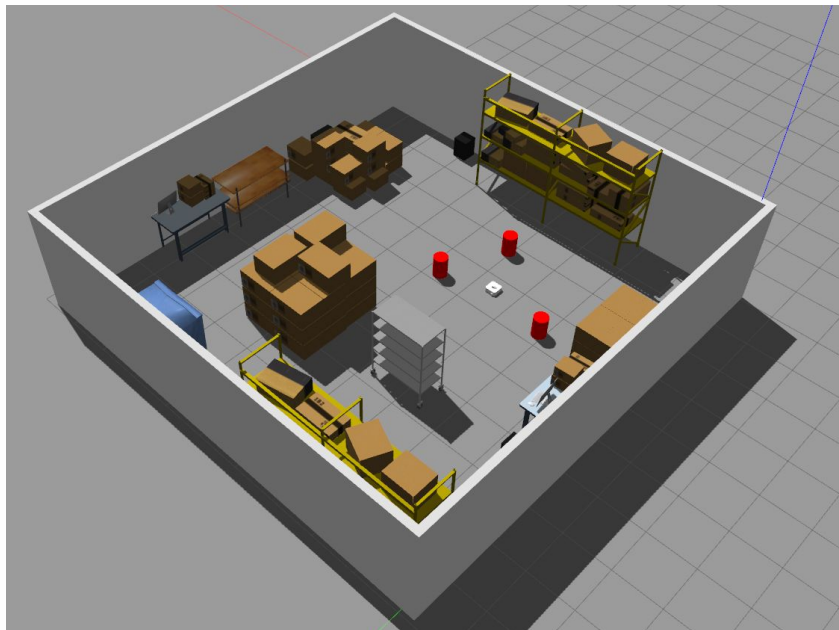
Stage 2: Path planning with LPA*

Dynamic environment (700 * 700 px),
known map which changes, point
robot





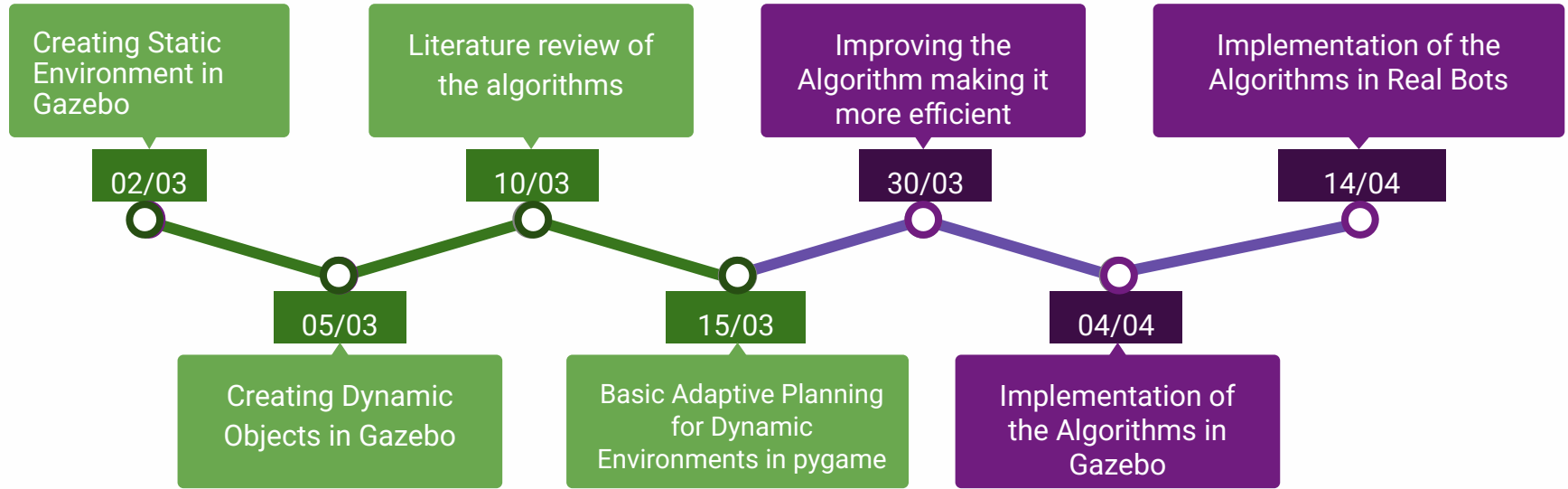
Environment in Gazebo



Proposal for dynamic path planning for RRT*

- The idea is similar to obstacle detection but the checking will happen with respect to time.
- Assumptions:
 - The obstacle is a wall that appears and disappears periodically.
 - The map and dynamic obstacles are known at all times. The velocity of the robot is constant.
 - Start with a point and holonomic robot, may be extended to a physical one at a later stage.
- Approach:
 - Since we know the current position of the bot, we can try expanding further.
 - Once we have the next point to move towards, we can first check for “spatial collision” with static obstacles.
 - If that is satisfied, we can check for “temporal collision”
 - Find time taken to travel from current point to next point and divide that into multiple segments
 - Predict the position of the robot and check for obstacle collision.
 - If there is a collision, sample again.
 - We can have a case where the robot can pass through the obstacle while it appears and disappears

MILESTONES AND PROGRESS



QUESTIONS!

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